One-stabilization irrationality and Hodge conservation for Fano threefolds

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Abstract

We prove that $X \times \mathbf{P}^1$ is irrational for every smooth complex cubic threefold X. More generally, for every smooth complex Fano threefold of Picard rank one, $X \times \mathbf{P}^1$ is rational if and only if X is rational. The proof constructs numerical birational invariants of fourfolds from selected generalized eigenspaces of quantum multiplication. The key step is to show that points, curves, and surfaces contribute zero to these invariants in the quantum blowup formula. Weak factorization then gives birational invariance, and quantum product calculations determine the invariants after product with $\mathbf{P}^1$. For the nine irrational families, a refinement retaining Hodge structures shows that birationality of $X \times \mathbf{P}^1$ and $Y \times \mathbf{P}^1$ forces an isomorphism of their rational third cohomology as Hodge structures. The numerical irrationality proofs are independent of this refinement.

Keywords. cubic threefold; universal CH_0-triviality; irrationality; quantum D-modules; formal exponents; birational invariants; weak factorization; Fano threefolds.

1 Introduction

We study whether irrationality survives product with one projective line. Stable rationality allows a projective-space factor of arbitrary dimension; here the dimension of the factor matters. Let X be a smooth complex cubic threefold. Clemens and Griffiths proved that X is irrational [11]. We prove that its first stabilization is irrational as well.

Theorem 1.1 (One-step irrationality)**.** *For every smooth complex cubic threefold X, the fourfold $X \times \mathbf{P}^1$ is irrational.*

The proof constructs an integer that is two on $X \times \mathbf{P}^1$ and zero on $\mathbf{P}^4$. It counts selected two-dimensional generalized eigenspaces of quantum multiplication. The essential step is to show that every point, curve and surface contributes zero to this count in the quantum blowup formula. Weak factorization then makes the count a birational invariant of fourfolds, proving the theorem.

Surface centers explain the difficulty. In the classical blowup formula, a surface S contributes $H^3(S)(-1)$ in degree five, where $H^3(X)(-1)$ also occurs in $H^5(X \times \mathbf{P}^1)$. The classical intermediate-Jacobian argument does not directly give the required obstruction. The quantum selection separates the cubic contribution from every possible center contribution.

The cubic calculation is the model case of a construction covering the entire classification of smooth Fano threefolds of Picard rank one. Here "one stabilization" always means product with $\mathbf{P}^1$.

Theorem 1.2 (Rationality after one stabilization)**.** *For every smooth complex Picard-rank-one Fano threefold X,*

$$X \times \mathbf{P}^1 \text{ is rational} \iff X \text{ is rational.}$$

The rational families are $\mathbf{P}^3$, the smooth quadric, index-two degrees $4, 5$, and index-one genera $7, 9, 10, 12$. The other nine families have irrational first stabilization.

Beyond detecting irrationality, we ask what the first stabilization remembers about the original threefold. A second result recovers its rational third cohomology for the nine families detected by the numerical argument, denoted by $\mathcal{F}$.

Theorem 1.3 (Rational Hodge conservation)**.** *Let X, Y be smooth complex Picard-rank-one Fano threefolds, each of index two and degree $1, 2, 3$, or of index one and genus $2, 3, 4, 5, 6, 8$. If $X \times \mathbf{P}^1$ and $Y \times \mathbf{P}^1$ are birational, then*

$$H^3(X, \mathbf{Q}) \cong H^3(Y, \mathbf{Q})$$

as rational Hodge structures.

Both theorems apply to every smooth member of the stated families. The retained Hodge structure can distinguish birational classes among irrational products. Combined with a separate rational Torelli theorem, it recovers a very general cubic or quartic from its first stabilization among smooth hypersurfaces of the same degree (Section 8). The Hodge isomorphism carries no assertion about integral lattices or polarizations. The numerical theorem is independent of this refinement.

The stable-rationality picture varies across the moduli space. Hassett and Tschinkel identified varieties birational to cubic threefolds as the exceptional Fano-threefold case not covered by their stable-irrationality theorem [14]. More recently, Engel, de Gaay Fortman, and Schreieder proved that a very general complex cubic threefold is not stably rational [13, Corollary 1.4], whereas Tschinkel and Zhang constructed explicit stably rational smooth cubic threefolds over $\mathbf{Q}$ [26, Theorem 1.3]. Theorem 1.1 applies uniformly in both situations: the first stabilization of those stably rational examples is irrational. The companion paper *Sharpness of irrationality after one stabilization for cubic threefolds* [25] proves that this bound is sharp by giving two explicit examples for which $X \times \mathbf{P}^2$ is rational.

The proof uses the quantum D-module (QDM): the cohomology module over the ring of Novikov and bulk parameters, equipped with its flat quantum connection [19]. The Novikov parameters record curve classes, while the bulk parameters are coordinates on cohomology. By the even QDM we mean the restriction to even cohomology with the odd bulk variables set to zero. We then replace the ring of numerical Novikov and even bulk parameters by its fraction field; this is what *generic* means below. In particular, it refers to quantum parameters, not to a general point in the moduli space of cubic threefolds. In the direction of the connection variable z, the leading coefficient is quantum multiplication by the Euler field. Over an algebraic closure of the generic coefficient field, its whole generalized eigenspaces, called *primary factors*, decompose the formal connection into blocks.

The count selects rank-two blocks for which the centered leading term N of Euler multiplication is nonzero and satisfies $N^2 = 0$. The lattice modification in Section 2 extracts two residue eigenvalues; flatness keeps their difference fixed as the quantum parameters vary. We count the block when the two residue eigenvalues are distinct modulo $\mathbf{Z}$, equivalently when the two formal exponent classes are distinct. Beauville's quantum product formulas give exponent representatives $-1/6$ and $-5/6$ for the cubic block, with difference $2/3$. These values already appear in Katzarkov–Lee–Svoboda–Petkov [1, pp. 375–376]; we give the residue calculation in our conventions. The small Gromov–Witten product formula and flatness in all even bulk directions show that it doubles after product with $\mathbf{P}^1$; see Lemma 4.2. This proves the cubic theorem.

This use of Euler eigenspace blocks in birational geometry belongs to the same circle of ideas as the Hodge-atom framework of Katzarkov–Kontsevich–Pantev–Yu [20]. Their framework combines the generalized eigenspaces of Euler multiplication, Iritani's blowup theorem, and weak factorization to produce birational invariants. In particular, their Example 6.21 uses the zero spectral piece, enhanced by the Serre operator and formal monodromy, to recover the irrationality of every smooth cubic threefold [20, Example 6.21]. Recent applications and refinements include work on cubic, Gushel–Mukai, and Küchle fourfolds [15, 9, 8].

For the one-stabilization problem, we specialize Iritani's blowup comparison so that the center summands remain distinct, and compute the product with one projective line from its small quantum connection [17, 30]. Gyenge gives a complementary formal tensor-product theorem for quantum D-modules of products and proves multiplicativity for Hodge atoms [16]. Cai also computes the same two exponent representatives and uses them with Iritani's blowup theorem to prove symplectic irrationality of cubic threefolds [10]; that argument does not address stabilization.

Keeping distinct eigenvalues of the canonical modified residue, even when their difference is integral, defines the stronger count I_{lat}. It detects index-two degree two and index-one genus six, where exponent classes alone do not distinguish the repeated factor. In genera two through five the repeated factor has even rank three, and its odd dimension gives the obstruction. Cai's quartic persistence theorem is a precedent for the cyclic-cluster argument [32, Theorem 5.3]. For the Hodge refinement we use Iritani's equivariant blowup comparison [31]. Cavenaghi–Katzarkov–Kontsevich already use atomic doubling under a $\mathbf{P}^1$ product to obstruct equivariant linearization, and retain curve-Jacobian isogeny classes in equivariant threefold invariants [2, Theorems E, F and H]. Benedetti–Guéré–Manivel–Perrin obtain rational-Hodge restrictions on cubic and Gushel–Mukai fourfolds birational to a Verra fourfold [3, Theorem 1]. Here the selected factors recover the whole rational H^3 for any smooth members of the nine stated families whose first stabilizations are birational.

The proof routes are separated as follows. The numerical results use neither Part II nor the optional general-bundle theorem.

Result	Required route
Cubic irrationality	Numerical invariant and center vanishing (Sections 2–3); cubic calculation and $\mathbf{P}^1$ product (Section 4).
Fano classification	Numerical route; rank-three odd dimension and family endpoints (Section 5); geometric matrix identifications (Appendix D).
Hodge conservation	Numerical selection and endpoints; equivariant fixed-base comparison and rational descent (Part II).

Readers following only the cubic route can stop after Section 4. The arithmetic application and the additive and general-bundle extensions are in the appendices. The latter extensions retain Iritani–Koto's theorem. Appendix B also explains the dimensional limits and the obstruction to extending this mechanism to a second stabilization.

Part I
Numerical obstruction

2 Numerical quantum invariants

The model is the cubic residue, with eigenvalues $-1/6, -5/6$, compared with the residue for a curve of genus at least two, whose eigenvalues coincide. The cubic count requires two eigenvalues distinct modulo integers; the stronger Fano count requires only that they be distinct. Both tests apply to rank-two factors with nonzero square-zero centered Euler operator. Section 3 extends vanishing to surfaces.

For a first pass, retain residue invariance (Proposition 2.5), the blowup formula (Proposition 3.2) and center vanishing (Proposition 3.4), then read the cubic calculation in Section 4.2 and the product Lemma 4.2. The persistence arguments justify using those small calculations in all even bulk directions.

2.1 Whole primary connection blocks

Let Y be smooth and projective. We first make three setup choices. Pass to numerical Novikov variables, set the odd bulk variables to zero, and restrict the underlying cohomology space to $H^{\mathrm{ev}}(Y)$. The last restriction is essential: restricting the bulk base alone would still leave odd cohomology in the fibre. Use graded completions as in [17, Section 2.2], and let K_Y be the fraction field of the numerical divisor-equation-reduced coefficient domain defined in Section 3. Fix an algebraic closure $\overline{K}_Y$. We form all spectral blocks only after base change to $\overline{K}_Y$, so no later scalar extension can split a factor that has already been counted as one block.

For a homogeneous even cohomology basis (ϕ_α), our conventions are

$$\nabla_{\partial_{t^\alpha}} = \partial_{t^\alpha} + z^{-1}(\phi_\alpha \star_t), \qquad \nabla_{z\partial_z} = z\partial_z - z^{-1}(E_Y \star_t) + \mu_Y,$$

where $n = \dim Y$, $\mu_Y|_{H^k(Y)} = (k-n)/2$, and

$$E_Y = c_1(Y) + \sum_\alpha \left(1 - \frac{\deg \phi_\alpha}{2}\right) t^\alpha \phi_\alpha.$$

Horizontal sections therefore satisfy $z^2\partial_z y = ((E_Y \star_t) - z\mu_Y)y$. Let U denote multiplication by the Euler field on $H^{\mathrm{ev}}(Y)$. For a primary block with eigenvalue λ, centering removes the scalar exponential factor $e^{-\lambda/z}$; its leading operator becomes $N = U - \lambda I$.

We always retain a whole generalized eigenspace, rather than individual Jordan blocks inside it. After base change to $\overline{K}_Y$, each generalized eigenspace of U is a summand of the full formal connection, not just an invariant subspace of U. The splitting uses a gauge, that is, a formal change of frame, regular at $z = 0$:

Proposition 2.1 (Formal spectral connection splitting)**.** *Suppose the leading operator of a finite free formal QDM has block-diagonal form $U_i = \lambda_i I + N_i$, where each N_i is nilpotent and $\lambda_i - \lambda_j$ is a unit for $i \neq j$. There is a unique gauge $G = I + O(z)$, regular in z and block off-diagonal in positive orders, which makes the z-connection block diagonal. In this gauge every bulk and Novikov direction is block diagonal as well. The horizontal pairing is block diagonal and nondegenerate on each factor.*

Proof. At order z^{n-1}, the (i,j)-entry of the gauge equation is a Sylvester equation

$$U_i X - X U_j = B_{n,ij}.$$

Its operator is $(\lambda_i - \lambda_j)I$ plus the nilpotent operator $X \mapsto N_i X - X N_j$, hence is invertible. Recursion gives the normalized gauge, and the same equation with zero right-hand side proves uniqueness.

Choose frames of the generalized-eigenspace subbundles after the splitting extension. If an off-diagonal block of a base-direction connection had a first nonzero Laurent coefficient Xz^m, the lowest off-diagonal coefficient of flatness would be $U_i X - X U_j = 0$, contrary to Sylvester invertibility. Thus every base direction preserves the same factors. Applied coefficientwise to pairing horizontality, the identical argument kills the off-diagonal pairing blocks. Nondegeneracy of the total pairing then makes every diagonal restriction nondegenerate. □

We call these factors the *generic even QDM blocks* of Y. A block retains its connection, pairing, and centered Euler endomorphism, but not an absolute cohomological grading. Its morphisms preserve these data and are regular connection isomorphisms after scalar extension and formal change of bulk coordinates. In particular, their gauges use neither negative nor fractional powers of the connection variable z.

2.2 Rank-two residues and two counts

Write E for the original $\overline{K}_Y[[z]]$-lattice of a generic even block: its module of formal sections regular at $z = 0$. Inverting z gives the meromorphic connection space. After centering, the equation is

$$z\partial_z y = A(z)y, \qquad A(z) = z^{-1}N + A_0 + zA_1 + z^2 A_2 + \cdots . \tag{2.1}$$

Assume $\operatorname{rk} E = 2$, $N \neq 0$, and $N^2 = 0$. Then $L = \operatorname{im} N = \ker N$ is a line. The following matrix calculation explains the modification. In a frame adapted to L, suppose that A_0 preserves this line, so that

$$N = \begin{pmatrix} 0 & \nu \\ 0 & 0 \end{pmatrix}, \qquad A_0 = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \qquad (A_1)_{21} = c.$$

Changing the frame by $S = \operatorname{diag}(1, z)$ removes the pole and gives residue

$$R = \begin{pmatrix} a & \nu \\ c & d-1 \end{pmatrix}.$$

The lower-left entry explains why the first-order coefficient A_1 is needed; the subtraction of one comes from differentiating the new frame. Eigenvalues $0, 1$ represent the same class modulo $\mathbf{Z}$ and do not qualify for I_{exp}, although they qualify for I_{lat}; eigenvalues $0, 2/3$ represent distinct classes.

The canonical elementary modification along this line is

$$E^\sharp = \{s \in E : s \bmod z \in L\}. \tag{2.2}$$

For this modification to remove the pole, the regular coefficient A_0 must preserve L. Pairing horizontality supplies exactly that condition.

Lemma 2.2 (The regular coefficient preserves the nilpotent line)**.** *Horizontality of the Poincaré pairing implies $A_0 L \subset L$.*

Proof. Separated spectral factors are orthogonal for the horizontal Poincaré pairing: coefficientwise horizontality kills every off-diagonal block by an invertible Sylvester operator.

Hence the restriction to E is nondegenerate. Write it as $P(z) = P_0 + zP_1 + \cdots$. Horizontality gives

$$z\partial_z P(z) + A(z)^T P(z) + P(z)A(-z) = 0.$$

The z^{-1} coefficient gives $N^T P_0 = P_0 N$. Thus L is isotropic and, in dimension two, $L^\perp = L$. The constant coefficient is

$$A_0^T P_0 + P_0 A_0 + N^T P_1 - P_1 N = 0.$$

Evaluating this identity on (x, x), where $x \in L = \ker N$, gives $(A_0 x, x) = 0$. Hence $A_0 x \in L^\perp = L$. □

Choose a frame with $L = \langle e_1 \rangle$ and $Ne_2 = \nu e_1$, $\nu \neq 0$. The modified lattice has frame e_1, ze_2, so

$$A^\sharp = S^{-1}AS - S^{-1}z\partial_z S, \qquad S = \mathrm{diag}(1, z). \tag{2.3}$$

The term $z^{-1}N$ becomes regular. The only possible new pole is the E_{21}-coefficient of A_0, which vanishes by Lemma 2.2. Thus $A^\sharp = R + zR_1 + \cdots$. Define

$$\delta^\sharp(E) := (\mathrm{tr}\, R)^2 - 4 \det R. \tag{2.4}$$

Lemma 2.3 (Persistence of a cyclic double-primary cluster)**.** *Over a characteristic-zero formal bulk germ, let a rank-two spectral cluster of a flat QDM be separated from its complement at the closed point. Suppose its centered leading operator specializes to a nonzero square-zero operator, and its scalar-centered base equations have at most a simple z-pole. The cluster remains cyclic and its centered leading operator remains nonzero square-zero throughout the germ.*

Proof. Write a centered base equation as $\partial y = (z^{-1}C_\partial + C_{\partial,0} + O(z))y$. The z^{-2} coefficient of flatness gives $[N, C_\partial] = 0$, and the trace of the z^{-1} coefficient gives $\mathrm{tr}\, C_\partial = 0$; here the base equation has been scalar-centered. Nilpotence of N away from the base point is not yet known, so the commutant is computed over the complete local ring $\mathcal{O}$ of the formal bulk germ. The rank-two spectral cluster continues over $\mathcal{O}$ because it is separated from the complementary spectral clusters at the base point; N is the centered z^{-1} coefficient of that continuation, so $\mathrm{tr}\, N = 0$, and it specializes at the closed point to a nonzero rank-one nilpotent. Choose v with v, Nv a basis modulo the maximal ideal; by Nakayama it is a basis over $\mathcal{O}$, so N is cyclic. Any C commuting with a cyclic N is determined by $Cv = \alpha v + \beta Nv$ and equals $\alpha I + \beta N$, whether or not N is nilpotent. Hence $[N, C_\partial] = 0$ and $\mathrm{tr}\, C_\partial = 0 = \mathrm{tr}\, N$ give $C_\partial = q_\partial N$ with $q_\partial \in \mathcal{O}$. The same z^{-1} coefficient gives

$$\partial N = -q_\partial N + [N, q_\partial A_0 - C_{\partial,0}].$$

Writing $B = q_\partial A_0 - C_{\partial,0}$, one has $\partial(N^2) = -2q_\partial N^2 + [N^2, B]$; uniqueness for the formal coefficient recursion preserves $N^2 = 0$. An entry nonzero at the closed point remains a unit, so N has rank one throughout the formal germ. □

The next lemma is needed only for the rank-three Fano families in Section 5. For the cubic theorem, continue with Proposition 2.5.

Lemma 2.4 (Persistence of a cyclic triple-primary cluster)**.** *Under the formal-germ and connection hypotheses of Lemma 2.3, a separated rank-three cluster whose centered leading operator is regular nilpotent at the closed point remains cyclic and nilpotent throughout all even bulk directions.*

Proof. Nakayama's lemma continues a cyclic frame v, Nv, N^2v. A commuting base pole is therefore $C = a_0I+a_1N+a_2N^2$. Put $p_j = \mathrm{tr}(N^j)$. Centering gives $p_1 = 0$ and $a_0 = -a_2p_2/3$. The same flatness coefficient as in the rank-two proof gives $\partial N = -C+$ commutators whose traces against powers of N vanish. Cayley–Hamilton gives $\mathrm{tr}(N^4) = p_2^2/2$, hence

$$\partial p_2 = -2a_1p_2 - 2a_2p_3, \qquad \partial p_3 = -3a_1p_3 - \tfrac{1}{2}a_2p_2^2.$$

The zero initial values have the unique zero solution by recursion in formal bulk degree, in characteristic zero. Thus the characteristic polynomial is T^3 and $N^3 = 0$. Cyclicity gives $N^2 \neq 0$. □

Proposition 2.5 (Invariance of the modified residue)**.** *Under formal change of even bulk coordinates, the modified connection is regular in every base direction and*

$$\partial R = [G_\partial, R] \tag{2.5}$$

for a regular matrix G_∂. Consequently $\partial(\delta^\sharp) = 0$. Regular connection isomorphisms preserve the domain and value of $\delta^\sharp$.

Proof. Lemma 2.3 preserves the nilpotent line. Its centralizer argument writes the centered leading base coefficient as $q_\partial N$. In a frame with $N = \nu E_{12}$, the modification $\mathrm{diag}(1, z)$ makes this upper-right pole regular. Only the lower-left entry of the regular base coefficient can then create a pole, of the form $z^{-1}kE_{21}$. If $R = \left(\begin{smallmatrix} a & \nu \\ c & d \end{smallmatrix}\right)$, the z^{-1} flatness equation is $kE_{21} + [R, kE_{21}] = 0$. Its diagonal entries are νk and $-\nu k$, so $k = 0$. The constant term of the flatness equation is now (2.5); trace and determinant are constant under infinitesimal conjugation. A regular isomorphism carries N, L, and the modified lattice to their counterparts and conjugates the residue. Indeed, if G, G^{-1} are regular on the original lattices, the induced map $S_{\mathrm{target}}^{-1}GS_{\mathrm{source}}$ and its inverse are regular on the modified lattices. Reduction modulo z gives the conjugating matrix; its lower-left entry can involve the first coefficient of G, so it need not be $G(0)$ in the original frames. □

Proposition 2.6 (Exponent interpretation)**.** *If ρ_1, ρ_2 are the eigenvalues of R, their classes modulo $\mathbf{Z}$ are the formal exponent classes of the original centered block and*

$$\delta^\sharp = (\rho_1 - \rho_2)^2.$$

Proof. The elementary modification uses integral powers of z, so it preserves formal exponent classes modulo $\mathbf{Z}$. The modified connection is regular singular with residue R; the regular-singular classification identifies the exponent classes with the residue eigenvalues modulo $\mathbf{Z}$ [24, Chapter 3]. The displayed identity is the quadratic discriminant formula. □

For such a block E, write ρ_1, ρ_2 for the eigenvalues of its modified residue in the fixed algebraic closure $\overline{K}$ of the generic coefficient field. Define the invariant on each block by

$$w_{\exp}(E) = \begin{cases} 1, & \mathrm{rk}\, E = 2,\ N \neq 0,\ N^2 = 0,\ [\rho_1] \neq [\rho_2] \text{ in } \overline{K}/\mathbf{Z}, \\ 0, & \text{otherwise}, \end{cases} \tag{2.6}$$

and put $I_{\exp}(Y) = \sum_E w_{\exp}(E)$. Call a block *eligible* if it has rank two and $N \neq 0$, $N^2 = 0$. Define the *lattice count* by

$$I_{\mathrm{lat}}(Y) = \#\{E \text{ eligible} : \delta^\sharp(E) \neq 0\}.$$

Nonzero discriminant means that the two eigenvalues of the canonical modified residue are distinct. The stronger test $\delta^\sharp \notin \{n^2 : n \in \mathbf{Z}\}$ means that their classes modulo $\mathbf{Z}$ are distinct. Thus $I_{\exp} \leq I_{\mathrm{lat}}$. Both counts are preserved by regular lattice isomorphisms and by common scalar shifts of the residue. For I_{lat}, preservation of the canonical modified lattice is essential: separate integral shifts of the two residue eigenvalues need not preserve their difference. Neither cyclic persistence nor residue conjugacy uses nonresonance.

3 Blowups and surface centers

The blowup formula needs three properties: faithful recovery of each intrinsic center coefficient field, transport of the original z-lattice, and separation of different center occurrences. Iritani's theorem supplies the graded-module connection isomorphism and inverse. Below we realize these maps as regular lattice maps, prove faithfulness using independent center divisor coordinates, and separate occurrences by their unit shifts. A first-pass reader may retain these conclusions and the formula in Proposition 3.2, then proceed to center vanishing. The original lattice is essential for the resonant rank-two cases.

The comparison and its lattice. Use Iritani's reduced graded source, its ambient and separate center images, and the independent external coordinates of Section 5.8.2 [17, Remarks 2.3 and 5.6, Theorem 5.18]. Their generic fields embed in a common algebraically closed overfield after the prescribed exceptional Novikov Laurent extension and ramification. This does not identify two differently directed Novikov completions. The comparison is parity preserving; on the even bulk base it and its inverse restrict to even cohomology. The same maps act on the full supermodule when odd dimensions are retained below.

Theorem 5.18 gives mutually inverse maps of the specified graded completed $\mathbf{C}[z]$-modules, intertwining the actual z-connections. Their positive coefficientwise realization is regular in z in both directions [17, Remark 1.5]. Hence they preserve the original lattices, and their leading terms conjugate N and $\operatorname{im} N$. They therefore preserve the elementary modification and conjugate its residue. A common grading normalization can add a scalar residue; it leaves the discriminant unchanged, including in the resonant case.

Faithful center coordinates. Iritani's center map need not be injective [17, (5.15)]. After numerical reduction it has the form

$$Q_Z^d \longmapsto Q^{i_* d} q_{\mathrm{exc}}^{-\rho_Z \cdot d/(c-1)}. \tag{3.1}$$

Here $\rho_Z = c_1(N_{Z/Y})$. The raw map is not the coefficient map used below. Remark 2.3 gives the intrinsic divisor-equation-reduced coefficient ring, whose Novikov generators are the combined symbols

$$X_d = Q_Z^d \exp\bigl(\sigma^{(2)} \cdot d\bigr);$$

the divisor coordinates $\sigma^{(2)}$ are not separate elements of this ring. Remark 5.6 places its pushforward under the raw Novikov map over the image of that reduced ring [17, Remarks 2.3 and 5.6]. In the external direct sum of Section 5.8.2, by contrast, the variables $s_j \in H^*(Z)$ are independent coordinates on the common comparison target, and the center parameter is identified as $\varsigma_j = \varsigma_j^\circ + s_j$ [17, (5.47)–(5.48)]. The divisor components of these independent target coordinates retain the curve-class information lost by (3.1).

The distinction is modeled by $x, y \mapsto q$, which identifies two generators, and $x \mapsto qe^s,\ y \mapsto qe^t$, which retains independent characters. Source coefficients must not already depend on the target variables s, t. The comparison uses the following conventions:

Object	Required data
Source	Reduced graded coefficients $\mathbf{C}[[X_d, \sigma']]_{\mathrm{gr}}[\sigma^0]$; no independent source divisor variables.
Target	Independent ambient and center-copy coordinates $t, s_0, \dots, s_{c-2}$; all center divisor directions retained.
Scalars	The displayed exceptional or fibre Novikov Laurent extensions and ramification; their z-free fraction fields in a common algebraic closure.
Connections	Graded completed z-enhancements embedded coefficientwise in the common field's $\overline{K}[[z]]$; comparison and inverse regular.
Derivations	Bulk partials, numerical Novikov logarithmic derivatives and the actual z-connection, with coordinate-change chain rules.
Grading	Only common scalar residue shifts; no independent shifts of residue eigenlines.

Lemma 3.1 (Faithful center base change)**.** *Let $R^{\mathrm{red}}_{Z,\mathrm{src}} = \mathbf{C}[[X_d, \sigma']]_{\mathrm{gr}}[\sigma^0]$ be the coefficient domain whose graded completed z-enhancement is Iritani's reduced source ring*

$$\mathbf{C}[z][[Q_Z e^{\sigma^{(2)}}, \sigma']][\sigma^0].$$

Let A_j be the numerical coefficient domain of the external direct sum in Section 5.8.2, with the independent target coordinates $t, s_0, \dots, s_{c-2}$ and with z omitted. The pullback determined by the center Novikov map and $\varsigma_j = \varsigma_j^\circ + s_j$ gives a continuous homomorphism

$$\iota_j : R^{\mathrm{red}}_{Z,\mathrm{src}} \longrightarrow A_j.$$

On a combined divisor-equation generator it is

$$X_d = Q_Z^d \exp\bigl(\sigma^{(2)} \cdot d\bigr) \longmapsto Q^{i_* d} q_{\mathrm{exc}}^{-\rho_Z \cdot d/(c-1)} \exp\bigl((\varsigma_j^{\circ,(2)} + s_j^{(2)}) \cdot d\bigr). \tag{3.2}$$

This map is injective. In particular, it induces a field embedding

$$\mathrm{Frac}(R^{\mathrm{red}}_{Z,\mathrm{src}}) \hookrightarrow \mathrm{Frac}(A_j).$$

After this field extension, each center summand in Iritani's comparison is the faithful scalar extension, followed by the formal coordinate change ς_j, of the intrinsic generic numerical even QDM of Z.

Proof. All completions here use the graded convention of [17, Section 2.2]. The pullback is defined on the reduced ring by the String and Divisor Equations, as explained after (5.36) and (5.47) there. It need not be defined on the unrestricted bulk power-series ring. We prove injectivity using the first nonzero Novikov degree, retaining the full exponentials in the target divisor coordinates. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(Z)$, and use their dual coordinates in $s_j^{(2)}$. The variables s_j occur only in the target A_j, not as coefficients in $R^{\mathrm{red}}_{Z,\mathrm{src}}$. This asymmetry is essential.

Filter by an integral ample degree pulled back from Y. Every fibre of the untagged monomial map in a bounded piece is finite: the restriction of an ample divisor from Y is ample on Z, and its bounded slice of $\overline{NE}(Z)$ contains only finitely many lattice points. This filtration is respected by the target map because $H \cdot i_* d = (i^* H) \cdot d$, and the fixed shifts have nonnegative Novikov degree.

For a fixed curve class, the coefficient of an element of the reduced ring is polynomial in the nondivisor and unit coordinates. Indeed, an element has finitely many homogeneous degrees and bounded unit degree; every other even nondivisor coordinate has strictly negative degree. Thus only finitely many of its monomials occur at that curve class. This is where the reduced, graded source is essential.

Suppose a nonzero source element maps to zero and take its first nonzero Novikov-degree part. At that degree only the Novikov-degree-zero part of ς_j° enters. Translation by its nondivisor and unit components is an injective substitution on the polynomial coefficients, over the target Laurent coefficient field. Its divisor component multiplies each exponential by a nonzero scalar. Group the terms by their common untagged target monomial. On one nonzero group the remaining relation has the form

$$\sum_{d \in F} a_d \exp\left(\sum_{k=1}^{\rho} (D_k \cdot d) s_{j,k}\right) = 0, \qquad a_d \neq 0,$$

over a characteristic-zero coefficient domain, including the remaining bulk coordinates, that does not contain the variables $s_{j,k}$. The exponent vectors are distinct. Choose an integral one-parameter restriction on which they remain distinct. Derivatives of orders $0, \dots, |F|-1$ at the origin give a Vandermonde matrix, hence every a_d is zero, a contradiction. This proves injectivity on the full reduced source. We have not truncated the divisor exponentials by bulk order; doing so could lose the distinction between curve classes.

Remark 2.3 of [17] gives the intrinsic reduced source, and Remark 5.6 gives its image under the raw Novikov map (5.15). Although that map can be noninjective on the unrestricted source, it retains the independent $\sigma^{(2)}$ coordinates in its target. Its restriction to the reduced source is injective by the same argument, with zero shift. Equations (5.47)–(5.48) then pull that image into the external direct sum over A_j, where the variables s_j are independent, and identify the direct sum with the blowup module. Associativity of scalar extension therefore identifies the center summand with scalar extension directly along the injective composite ι_j. Passing to fraction fields is faithfully flat. Finally, the inverse of the combined formal coordinate change $(\tau, \varsigma_0, \dots, \varsigma_{c-2})$, supplied by Theorem 5.18(7) and Section 5.8.2, transports the same statement to the original blowup coordinates. □

Proposition 3.2 (Blowup formula)**.** *For $I = I_{\mathrm{exp}}$ or I_{lat}, a smooth center $Z \subset Y$ of codimension $c \geq 2$ satisfies*

$$I(\mathrm{Bl}_Z Y) = I(Y) + (c-1)I(Z). \tag{3.3}$$

For a disconnected center the correction is summed componentwise.

Proof. Iritani's comparison has one ambient and $c-1$ center copies. Its joint coordinate map has an inverse, so their unit coordinates are independent. The resultant of two scalar-translated characteristic polynomials is a nonzero polynomial in the difference of the unit shifts. Thus their spectra are disjoint generically. Regular lattice transport preserves each selection condition, and Lemma 3.1 identifies each center copy with its intrinsic generic connection. Contributions therefore add. Each operation uses its own common field; no successive Laurent expansions are composed. □

Lemma 3.3 (The ruled product in full even bulk)**.** *For a smooth projective curve C, the two counts vanish on $C \times \mathbf{P}^1$. Its generic even primary factors have rank at most two. If $g(C) = 1$, every rank-two centered operator is zero; if $g(C) > 1$, every rank-two canonical residue discriminant is zero.*

Proof. For $g = g(C) \geq 1$, use the flat even basis $1, h, p, hp$, with h and p the point classes of the factors and $\int hp = 1$. Write the bulk as $v\,1 + ah + bp + chp$, set $Q = q_f e^b$ and $\kappa = 2-2g$. The horizontal Novikov variable remains an independent spectator. Every genus-zero map to C is constant. The positive-degree moduli space is C times the corresponding $\mathbf{P}^1$ moduli space, with no H^1 obstruction from the constant factor. A nonzero invariant has exactly one h factor. An insertion h alone gives a unit on the second factor and vanishes by the string equation. Otherwise there is one hp insertion; all other nonunit insertions are p, and the dimension equation forces fiber degree one. Its three-point invariant is one; the divisor equation handles the remaining p insertions. The full even potential, modulo terms of degree at most two, is therefore

$$F = \tfrac{1}{2}v^2c + vab + cQ.$$

Its third derivatives give $h \star h = 0$, $h \star p = hp$ and $p \star p = Q(1 + ch)$. The centered Euler element is $\kappa h + 2p - chp$. Put $p' = (1 - ch/2) \star p$. Then $(p')^{\star 2} = Q$ and $U = \kappa h + 2p'$. After adjoining $\sqrt{Q}$, the two projectors $(1 \pm p'/\sqrt{Q})/2$ have even rank two, with centered operator κh throughout the full even bulk. For $g = 1$ this is identically zero. For $g > 1$ it has rank one; at the small point the product connection is the tensor connection, whose curve factor has modified residue $\kappa E_{21} - \frac{1}{2}I$. Regular rank-one separation of the $\mathbf{P}^1$ factor changes this residue only by a scalar. Proposition 2.5 keeps its discriminant zero in all bulk directions. For $C = \mathbf{P}^1$, the small product has four distinct eigenvalues over the independent Novikov field; its rank-one factors persist in all even bulk directions by Proposition 2.1. □

3.1 Vanishing on centers through dimension two

Proposition 3.4 (Vanishing on low-dimensional centers). *For every smooth projective Z of dimension at most two, $I_{\mathrm{exp}}(Z) = I_{\mathrm{lat}}(Z) = 0$. The same vanishing holds for every center copy in a quantum comparison.*

Proof. It suffices to prove vanishing for I_{lat}, since $I_{\mathrm{exp}} \le I_{\mathrm{lat}}$. *Points and curves.* A point has one rank-one block. For $\mathbf{P}^1$, the relation $p \star p = Qe^{t_p}$ gives two rank-one generic blocks.

Let C have genus $g \ge 1$. There is no nonconstant genus-zero map to C, so its even big quantum product is classical. With $\int_C p = 1$ and $a = 2 - 2g$, its centered even connection is

$$z\partial_z Y = \left[z^{-1}aE_{21} + \mathrm{diag}(1/2, -1/2)\right] Y.$$

For $g = 1$, $N = 0$. For $g > 1$, the modification $S = \mathrm{diag}(z, 1)$ gives residue $aE_{21} - \frac{1}{2}I$, of discriminant zero. Its two eigenvalues, and therefore its two exponent classes, are equal. Thus every curve has zero invariant.

Surfaces with nef canonical class. Let S be a surface with K_S nef. For a quantum-product term with input degree b, non-scalar Euler degree $a \ge 2$, curve class d, and nonunit even bulk degrees $e_\nu \ge 2$, the dimension axiom gives output degree

$$a + b - 2c_1(S) \cdot d + \sum_\nu (e_\nu - 2) \ge b + 2.$$

The centered Euler operator strictly raises ordinary degree. Its only block is the whole even cohomology, of rank $b_0 + b_2 + b_4 \ge 3$, so its invariant is zero.

Other surfaces. A minimal surface with non-nef canonical class is $\mathbf{P}^2$ or a ruled surface $\mathbf{P}_C(V)$ [7, Chapter VI]. The relation $H^{\star 3} = Q$ makes $\mathbf{P}^2$ generically semisimple. A ruled surface is birational to $C \times \mathbf{P}^1$: its defining vector bundle is trivial over the generic point of C. Lemma 3.3 gives zero on that product. Projective surface factorization has only point centers, whose contributions in (3.3) are zero. Thus every ruled surface has zero count. Every nonminimal surface is obtained from a minimal one by point blowups; formula (3.3) adds only zero point terms. This proves intrinsic vanishing for every surface. Lemma 3.1 identifies every separate center summand with a faithful scalar extension of the corresponding intrinsic generic block, and the invariant is unchanged by that extension. □

4 The cubic calculation and one stabilization

4.1 A rational rank-two calculation

The zero-primary block can be separated without splitting the complementary eigenlines. This gives one calculation for three Fano families. On a first pass, retain formula (4.1) and proceed to the cubic in Section 4.2.

Proposition 4.1 (A universal modified residue)**.** *For $s = 2a + b$, $sq \neq 0$, consider*

$$z^2\partial_z y = (U_{a,b} + zD)y, \qquad U_{a,b} = \begin{pmatrix} 0 & aq & 0 & a^2q^2 \\ 1 & 0 & bq & 0 \\ 0 & 1 & 0 & aq \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad D = \tfrac{1}{2}\operatorname{diag}(3, 1, -1, -3).$$

Its characteristic polynomial is $T^2(T^2 - sq)$. Its zero-primary block has rank two, nonzero square-zero leading term and canonical modified residue

$$R_{a,b} = \begin{pmatrix} -(2a+3b)/(2s) & 1 \\ -4a^2/s^2 & (b-2a)/(2s) \end{pmatrix}, \qquad \delta^\sharp(a,b) = \frac{4(b-2a)}{b+2a}. \tag{4.1}$$

Here a, b, q are constant under ∂_z.

Proof. The columns of

$$C = \begin{pmatrix} aq & 0 & 0 & -(a+b)q \\ 0 & (a+b)q & -aq & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \qquad \det C = -s^2q^2,$$

give $\operatorname{im} U^2 \oplus \ker U^2$. Direct multiplication gives

$$J = C^{-1}UC = J_+ \oplus N, \qquad J_+ = \begin{pmatrix} 0 & sq \\ 1 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Put $B = C^{-1}DC$. The normalized gauge $G = I + zX + z^2Y + \cdots$, with zero diagonal blocks in positive orders, starts with

$$X_{+0} = \operatorname{diag}(2a/s, 2/(qs)), \qquad X_{0+} = \operatorname{diag}(-2/(qs), -2a/s).$$

These matrices solve $J_iX_{ij} - X_{ij}J_j = -B_{ij}$. Writing the transformed coefficient as $J + zB_1 + z^2B_2 + \cdots$, its gauge equation gives

$$B_1 = B + [J, X], \qquad B_2 = BX - XB_1 - X + [J, Y].$$

The derivative contributes $-X$. Its diagonal blocks vanish, as do those of XB_1 and $[J, Y]$, so

$$(B_1)_{00} = \operatorname{diag}\left(-\frac{2a+3b}{2s}, \frac{2a+3b}{2s}\right), \qquad ((B_2)_{00})_{21} = (B_{0+}X_{+0})_{21} = -\frac{4a^2}{s^2}.$$

Modification by $\operatorname{diag}(1, z)$ gives $R_{a,b}$. Its characteristic polynomial is $T^2+T+(10a-3b)/(4s)$, proving the discriminant formula. For an original z^2F term, add $(C^{-1}FC)_{ii}$ to $(B_2)_{ii}$. Thus only the first off-diagonal gauge coefficient is needed. $\square$

4.2 The cubic block

Only a rank-two block can contribute to I_{exp}. For a cubic threefold, the calculation below isolates the unique such candidate and then computes the residue of its elementary modification.

Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold, let $P = H|_X$, and put $q = Q^\ell$ for the line class. Beauville's formulas give [6, main theorem and (2.1)–(2.3)]

$$P \star P = P^2 + 6q, \qquad P \star P^2 = P^3 + 15qP, \qquad P \star P^3 = 6qP^2 + 36q^2. \tag{4.2}$$

Thus, in the basis $(1, P, P^2, P^3)$,

$$U_X = 2\begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad D_X = \frac{1}{2}\operatorname{diag}(3, 1, -1, -3), \tag{4.3}$$

and $z^2\partial_z y = (U_X + zD_X)y$. This agrees with [10, Section 3].

The change of basis $\operatorname{diag}(1, 2, 4, 8)$ changes $U_X = 2U_{6,15}$ to $U_{24,60}$ and leaves D_X unchanged. The universal calculation therefore applies. Conjugating its residue by $\operatorname{diag}(2, 1)$ gives the original cubic normalization:

$$R_X = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}. \tag{4.4}$$

Its characteristic polynomial is $(T + 1/6)(T + 5/6)$. Thus the two exponent classes differ by $2/3 \notin \mathbf{Z}$, so the block is counted. Equivalently,

$$\boxed{\delta^\sharp(X) = \frac{4}{9}}, \qquad I_{\mathrm{exp}}(X) = 1. \tag{4.5}$$

From the small connection to the generic block. After inverting q, at the small quantum point the rank-two cluster is separated from the other eigenvalues. The cyclic-centralizer argument in Lemma 2.3 continues that cluster over the formal bulk ring and proves that its centered leading term remains nonzero and square-zero. The modified residue is then preserved by flatness. Thus the small calculation determines the exponent count at generic quantum parameters, for every smooth cubic threefold.

Lemma 4.2 (Product with one projective line)**.** *Let T be a smooth member of the nine families $\mathcal{F}$ listed in Theorem 1.3. Suppose at the small quantum point over the generic Novikov field its even Euler operator has separated cyclic primary factors of ranks at most three. On $T \times \mathbf{P}^1$ each such factor occurs twice with the same even rank and cyclic centered operator in the full even bulk germ. A rank-two factor keeps its canonical modified residue discriminant. Consequently both rank-two counts double.*

Proof. The small genus-zero product formula identifies the full connection and original lattice with the tensor product: Euler multiplication and grading are sums of those on the factors [30, formula (1)]. The two $\mathbf{P}^1$ eigenvalues are $\pm 2\sqrt{q_f}$. All shifted T clusters are distinct, since equality between opposite shifts would make the independent q_f algebraic over the coefficient field of T. The regular splitting of each rank-one $\mathbf{P}^1$ factor preserves its original lattice. Tensoring with it adds only scalar connection coefficients, so rank-two residue discriminants are unchanged. Lemmas 2.3 and 2.4 continue each cyclic cluster in all even bulk coordinates, including the mixed classes $\alpha \otimes p$. Proposition 2.5 continues the discriminant. This uses flatness in the transverse directions; a big quantum tensor formula at mixed bulk is not required. □

4.3 The projective endpoint and the contradiction

Lemma 4.2 gives

$$I_{\exp}(X \times \mathbf{P}^1) = 2. \tag{4.6}$$

On the other hand, over the generic numerical Novikov field $\Lambda = \operatorname{Frac} \mathbf{C}[[Q]]$, $QH^\bullet(\mathbf{P}^4) = \Lambda[H]/(H^5 - Q)$. At $Q \neq 0$, multiplication by $5H$ has five distinct eigenvalues after algebraic closure. Thus

$$I_{\exp}(\mathbf{P}^4) = 0. \tag{4.7}$$

Proof of Theorem 1.1. Proposition 3.4 makes every actual center correction zero in a weak factorization of smooth projective fourfolds. Hence projective weak factorization and (3.3) make $I_{\exp}$ birationally invariant. If $X \times \mathbf{P}^1$ were rational, then

$$2 = I_{\exp}(X \times \mathbf{P}^1) = I_{\exp}(\mathbf{P}^4) = 0,$$

a contradiction. □

5 Picard-rank-one Fano threefolds

The cubic is the rank-two model for a larger numerical obstruction. Five families are detected by the canonical rank-two residue. Four more have a repeated factor of even rank three; there we count its odd cohomology. No Hodge group is needed for either construction.

A resonant example: degree two. For an index-two Fano threefold of degree two, the small matrix in Appendix D has $a = 48$, $b = 160$ in (4.1). Substitution gives

$$R_{48,160} = \begin{pmatrix} -9/8 & 1 \\ -9/64 & 1/8 \end{pmatrix}, \qquad \det(TI - R_{48,160}) = T(T+1).$$

The residue eigenvalues are $0, -1$. Their exponent classes modulo $\mathbf{Z}$ coincide, so this block contributes zero to $I_{\exp}$. The canonical lattice retains their nonzero difference: $\delta^\sharp = 1$, so it contributes one to I_{lat}. The rank-one complements contribute nothing. This is why retaining the lattice detects a family that the exponent count misses.

5.1 Odd dimension on rank-three factors

Use the full quantum cohomology supermodule over the same even bulk base. The even Frobenius algebra acts on it. Its spectral projectors split both parities, and the full even rank determines the primary factor. Define

$$O_3(T) = \sum_{\operatorname{rk} E^{\mathrm{ev}} = 3} \dim E^{\mathrm{odd}},$$

where the sum runs over whole primary factors. We use the full odd dimension, so no divisibility convention is needed.

Lemma 5.1 (Odd dimension and surface vanishing)**.** *An even-rank-one primary factor of a Frobenius superalgebra has zero odd part. The count O_3 satisfies*

$$O_3(\operatorname{Bl}_Z T) = O_3(T) + (c-1) O_3(Z)$$

for a smooth codimension-c center. It vanishes in dimensions at most two and is a birational invariant of smooth projective fourfolds.

Proof. Let e be the unit of an even-rank-one factor. For odd a, b write $ab = \gamma e$. Supercommutativity gives $(ab)^2 = 0$, hence $\gamma = 0$. The odd Frobenius pairing is therefore zero; nondegeneracy forces the odd part to be zero.

The blowup comparison and its inverse preserve parity on the full fiber. They preserve full even ranks; independent occurrence-unit shifts separate their primary factors as in Proposition 3.2. The odd dimensions consequently add, and Tate suspensions preserve parity. Points and curves have even rank at most two. The ruled-product computation in Lemma 3.3 has no even-rank-three factor. Point blowups and surface factorization extend this to all ruled surfaces. Projective space has simple generic factors.

For a minimal surface with nef canonical class, the degree argument in Proposition 3.4 gives one whole even factor of rank b_2+2. It is selected only if $b_2 = 1$. In that case a nonzero holomorphic one-form α would give a nonzero real class $[i\alpha \wedge \bar{\alpha}]$ of square zero: its product with a Kähler class has positive integral. This is impossible when H^2 is generated by an ample class. Thus $H^1 = 0$ and, by duality, $H^3 = 0$. The selected odd dimension is zero. Surface classification and point blowups finish the vanishing proof. Weak factorization in dimension four then has zero correction at every step. □

5.2 The nine detected families

For index two put $d = H^3$, where $-K_X = 2H$. For index one put $g = \frac{1}{2}(-K_X)^3 + 1$. Let $\mathcal{F}$ be the nine families in the following table. The even cohomology of each member has rank four, and its only odd cohomology is H^3.

Proposition 5.2 (Fano endpoint data)**.** *Every smooth member of the indicated family has the following repeated small even primary factor; all complementary factors have rank one. The repeated factor is cyclic. The last two columns give the generic counts after continuation.*

Family	*Repeated even rank*	$\delta^\sharp$	I_{lat}	O_3
Index two, $d = 1$	*2*	16/9	*1*	*0*
Index two, $d = 2$	*2*	1	*1*	*0*
Index two, $d = 3$	*2*	4/9	*1*	*0*
Index one, $g = 2$	*3*	—	*0*	*104*
Index one, $g = 3$	*3*	—	*0*	*60*
Index one, $g = 4$	*3*	—	*0*	*40*
Index one, $g = 5$	*3*	—	*0*	*28*
Index one, $g = 6$	*2*	1	*1*	*0*
Index one, $g = 8$	*2*	4/9	*1*	*0*

Proof. Appendix D gives the nine matrices, their source normalizations and the exact finite calculation. The degree-one, two and three residue computation is also Proposition 4.1. The family formulas apply to every smooth member by smooth deformation invariance of Gromov–Witten invariants, including the special quartic and Gushel–Mukai presentations discussed in that appendix. The general Fano cohomological shape and the positive H^3 dimensions in genera two through five are those of the smooth classification [36, Introduction and Table 5]. Lemmas 2.3 and 2.4 continue the repeated factors in all even bulk directions; Proposition 2.5 continues the rank-two residues. By Lemma 5.1, no rank-one complement carries odd cohomology. Hence all of H^3 belongs to the repeated factor, giving O_3. Only its positivity is needed for irrationality. □

Proof of Theorem 1.2. The smooth classification has seventeen families [36, Introduction]. The eight families outside $\mathcal{F}$ are rational: $\mathbf{P}^3$, the smooth quadric, index-two degrees four and five, and index-one genera seven, nine, ten and twelve [23, Section 1.1]. Their products with $\mathbf{P}^1$ are rational.

For each of the remaining nine families, Proposition 5.2 gives $I_{\text{lat}} > 0$ or $O_3 > 0$. Lemma 4.2 doubles the former. For the latter, its tensor-product proof doubles the odd dimension at the small point; a continued projector has constant rank on a formal germ, so the same value holds in all even bulk directions. Both counts vanish on surface centers and on $\mathbf{P}^4$, and are birationally invariant in dimension four. Thus these nine products are irrational. A rational threefold has rational product with $\mathbf{P}^1$, proving the reverse implication as well. No deformation invariance of rationality is used. □

6 Consequences

We record the index-two specialization and the genus-eight correspondence, then combine the cubic theorem with criteria for universal CH_0-triviality.

6.1 Index-two Fano threefolds

Theorem 6.1 (Rationality after one stabilization in index two)**.** *Let V be a smooth complex Fano threefold with Picard rank one and $-K_V = 2H$, where H generates* $\mathrm{Pic}(V)$*. Then*

$$V \times \mathbf{P}^1 \text{ is rational} \iff V \text{ is rational} \iff H^3 \in \{4, 5\}.$$

Proof. Apply Theorem 1.2 to the five index-two families. The table in Proposition 5.2 gives the three nonzero residue discriminants; degrees four and five are rational [23, Section 2.3, Theorems 2.7, 3.3 and 3.5]. □

The negative conclusions also hold for forms over any characteristic-zero field. Indeed, a hypothetical rationalization and its inverse descend with the variety to a finitely generated field, which embeds into $\mathbf{C}$. For degree-four forms, rationality requires separate arithmetic hypotheses. Every smooth degree-five form is rational over its ground field [23, Theorem 3.3].

6.2 Birational partners and zero-cycles

Kuznetsov's associated Pfaffian cubic and flop supply the next application. Tschinkel and Zhang study equivariant and twisted forms of this correspondence [27].

Corollary 6.2 (Irrationality after one stabilization for V_{14})**.** *For every smooth prime Fano threefold V of genus eight, $V \times \mathbf{P}^1$ is irrational.*

Proof of Corollary 6.2. Kuznetsov's flop gives a birational map between rank-two projective bundles $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$ over V and its associated Pfaffian cubic threefold X [21, Theorems 2.17–2.18]. Each projective bundle is birational to the product of its base with $\mathbf{P}^1$. Hence $V \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$, which is irrational by Theorem 1.1. □

Corollary 6.3 (Separation in codimension at most three)**.** *There is a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds such that, for every member X, both X and $X \times \mathbf{P}^1$ are universally CH_0-trivial, while $X \times \mathbf{P}^1$ is irrational.*

Proof of Corollary 6.3. Voisin produces a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds along which the primitive minimal class $\Theta^4/4!$ is algebraic, hence along which CH_0 is universally trivial [28, Theorem 4.5 and Lemma 4.6]. The projective-bundle formula passes universal CH_0-triviality to the product, and Theorem 1.1 supplies its irrationality. □

Corollary 6.4 (Fermat cubic)**.** *Let*

$$X = \{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \subset \mathbf{P}^4.$$

Then the fourfold $X \times \mathbf{P}^1$ is universally CH_0-trivial and irrational.

Proof of Corollary 6.4. The Fermat equation is a sum of five forms in separated single variables, so [12, Theorem 2.8] gives universal CH_0-triviality; the projective-bundle formula passes it to the product. Irrationality follows from Theorem 1.1. □

Corollary 6.5 (Coprime-degree cubics)**.** *Let X be one of the cubic threefolds of [29, Theorem 3.3]. Then $X \times \mathbf{P}^1$ admits unirational parametrizations of coprime degrees, is universally CH_0-trivial, and is irrational.*

Proof of Corollary 6.5. The cubics of [29, Theorem 3.3] carry unirational parametrizations of degrees two and three, and [29, Corollary 3.5] keeps both degrees on $X \times \mathbf{P}^1$. Parametrizations of coprime degrees give a decomposition of the diagonal and hence universal CH_0-triviality [29, Section 1]. Theorem 1.1 gives irrationality. □

In dimension five, threefold centers can contribute to weak factorization. Appendix B retains those contributions and places the blowup formulas in the additive Grothendieck group.

Part II
Hodge conservation

7 Hodge conservation after one stabilization

For the nine families $\mathcal{F}$ detected in Part I, we refine the numerical invariant to recover two copies of $H^3(X, \mathbf{Q})$ from $X \times \mathbf{P}^1$. Points, curves and surfaces still contribute zero. Birational first stabilizations therefore give equal Hodge representation classes; cancellation and rational descent yield Theorem 1.3.

The new point is to restrict the quantum parameters while retaining the entire cohomology fiber. On this restricted base we check the blowup comparison, then reuse the block selection, surface vanishing and product calculations from Part I. The numerical classification is independent of this refinement.

7.1 A common group and the retained fiber

A common group lets us compare the Hodge structures of every variety and center in a weak factorization as representations of the same group. For a polarizable rational Hodge structure, the Hodge group is the rational algebraic group generated by the circle action that acts on $H^{p,q}$ as λ^{p-q}. Tate twists do not change this action. Iritani recalls its universal form in [31, Section 3]: the category of polarizable rational Hodge structures modulo Tate twists has a rational fiber functor, and its tensor automorphisms form the universal Hodge group.

Fix the finite collection of varieties and centers in a projective weak factorization of the proposed birational map, together with the auxiliary surface models used below. Use one reductive rational quotient G of this universal group through which all their cohomology representations factor. The image on each cohomology space is its Hodge group. Thus the choices for different varieties are compatible. Tate suspensions in the blowup formula act trivially on these representations.

For each variety T, restrict the bulk parameter to

$$H^*(T, \mathbf{Q})^G \otimes_{\mathbf{Q}} \mathbf{C}, \qquad \text{while the fiber remains } H^*(T, \mathbf{Q}) \otimes_{\mathbf{Q}} \mathbf{C}.$$

The fixed parameters are even: the element -1 of the Hodge circle acts by cohomological parity. The unit and algebraic divisor classes are fixed. No assertion that every Hodge class is algebraic is needed. Novikov parameters, z and the coefficient ramification parameters have trivial G-action and even parity.

The quantum product is equivariant under simultaneous action on parameters and fibers [31, Lemma 7]. On this fixed base its Euler operator therefore commutes with G. Its whole primary projectors commute with G as well. All ranks below are full even fiber ranks, never dimensions of G-invariant subspaces.

After a splitting extension of the generic coefficient field, record the odd part of each primary factor by its multiplicities in the irreducible representations of $G_{\mathbf{C}}$. These give a class in the representation group $R(G_{\mathbf{C}})$, a free abelian group on simple representations. Further algebraically closed coefficient extension does not change these multiplicities.

7.2 Faithfulness on the fixed base

The comparison must recover the intrinsic generic center connection, even when its raw Novikov map identifies curve classes. We check faithfulness on the fixed base directly. The new check is that algebraic divisor coordinates are fixed, so they still recover the curve-class information.

Lemma 7.1 (Fixed-base comparison)**.** *For a blowup $\mathrm{Bl}_Z T \to T$ with smooth connected center of codimension $c \geq 2$, the actual comparison on Hodge-fixed bulk bases preserves whole primary factors, their full G-representations and their original z-lattices. Each center occurrence is a faithful extension of its intrinsic reduced fixed-base connection. Eligible rank-two factors retain their canonical modified residue discriminant.*

Proof. Iritani's blowup isomorphism intertwines the quantum connections over the specified graded completed $\mathbf{C}[z]$-modules [17, Theorem 5.18 and Section 5.8.2]. Its coordinate maps and full-fiber isomorphism are equivariant, and their initial shifts are fixed [31, Proposition 8 and Lemma 10]. The joint coordinate map and its inverse restrict to inverse maps of fixed loci. In the external direct sum the fixed coordinates of the ambient copy and each center copy are independent; in particular their unit and divisor coordinates survive.

Write d for an effective integral numerical curve class of Z. The reduced source retains the combined generator $X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d)$, with no separate source divisor variables. If u denotes fixed higher-degree coordinates and v the unit coordinate, its ring is

$$R_Z^G = \mathbf{C}[[X_d, u]]_{\mathrm{gr}}[v].$$

Here graded completion retains only finitely many total homogeneous degrees in each element, and unit dependence is polynomial. At a fixed curve class the coefficient is consequently polynomial in u, v: every u has strictly negative degree. These are precisely the reduced-domain conventions of the numerical comparison proof.

For the jth occurrence, with independent fixed divisor coordinate $s_j^{(2)}$ and fixed initial shift ς_j°, the map is

$$X_d \longmapsto Q^{i_* d} q^{-c_1(N_{Z/T})\cdot d/(c-1)} \exp\big((\varsigma_j^{\circ,(2)} + s_j^{(2)}) \cdot d\big).$$

The other coordinates undergo the corresponding fixed shifts. Existence of this reduced substitution uses the string and divisor equations; it does not authorize translation of unrestricted formal power series.

To prove injection, take the least nonzero degree with respect to the restriction of an ambient ample divisor. Bounded numerical effective-degree slices are finite. At this least degree, positive-degree corrections to the initial shifts cannot contribute; translation of the polynomial coefficients by their degree-zero shifts is injective. Group terms with the same untagged target monomial. A kernel element would give a nonzero finite relation

$$\sum_d a_d \exp\left(\sum_k (D_k \cdot d) x_k\right) = 0,$$

where the a_d are independent of the divisor coordinates x_k. Choose integral algebraic divisors D_k separating numerical curve classes. Their coordinates remain available on the fixed base. Restrict to an integral line in these coordinates separating the finitely many exponent vectors. Successive derivatives at zero give an invertible Vandermonde matrix, forcing all $a_d = 0$. This is a contradiction.

The original comparison and its inverse are regular at $z = 0$ in the coefficientwise realization of the graded completion. Leading-term intertwining therefore carries N and $\operatorname{im} N$ to their counterparts, and hence carries the canonical modification

$$E^\sharp = \{s \in E : s \bmod z \in \operatorname{im} N\}$$

to the corresponding modification. In modified frames the regular inverse still exists, so the residues are conjugate. This applies also to resonant rank-two factors. Finally, independent occurrence-unit shifts separate their spectra generically: the resultant of two scalar-translated characteristic polynomials is a nonzero polynomial in the shift difference. Thus distinct occurrences do not merge in the comparison. □

The same argument applies to the ambient copy and componentwise to a disconnected center. Each operation uses its own comparison field; the resulting equalities of representation multiplicities telescope without composing different Laurent expansions along the factorization.

7.3 Selection and vanishing

Let E run over the whole primary factors of the full connection on the fixed base. Retain E if either

$$\operatorname{rk} E^{\mathrm{ev}} = 2, \quad N_E^2 = 0 \neq N_E, \quad \delta^\sharp(E) \neq 0, \qquad \text{or} \qquad \operatorname{rk} E^{\mathrm{ev}} = 3.$$

Here N_E is the centered leading operator on the even part, and $\delta^\sharp$ is the canonical rank-two residue discriminant. Define

$$\mathcal{H}_G(T) = \sum_{E \text{ retained}} [E^{\mathrm{odd}}] \in R(G_{\mathbf{C}}).$$

There is no factor of one half. Selection precedes extraction of the odd representation. The sum has no scalar eigenvalue or discriminant labels; permuting splitting branches does not change it. We do not require each individually labelled factor to descend to $\mathbf{Q}$.

Proposition 7.2 (Operation and vanishing)**.** *The selected class satisfies*

$$\mathcal{H}_G(\mathrm{Bl}_Z T) = \mathcal{H}_G(T) + (c-1)\mathcal{H}_G(Z).$$

It is zero for smooth projective varieties of dimension at most two and is unchanged under birational maps of smooth projective fourfolds.

Proof. The fixed-base comparison preserves the selection conditions and the odd representations, with the $c-1$ center Tate suspensions forgotten by G. Generic separation of occurrences makes their contributions add.

The low-dimensional calculations of Propositions 3.4 and Lemma 5.1 apply to these bases as follows. A point has only an even-rank-one factor. A curve has simple even factors, zero centered operator, or a rank-two factor with discriminant zero. For a minimal surface with nef canonical class, the centered Euler operator strictly raises ordinary degree, also on the fixed base. Its only primary factor has full even rank b_2+2. The only possible selected rank is three, when $b_2 = 1$. In that case a nonzero holomorphic one-form α would give a nonzero class $[i\alpha \wedge \bar{\alpha}]$ of square zero, impossible in the one-dimensional second cohomology generated by an ample class. Thus H^1 and H^3 vanish, and its odd contribution is zero.

On $C \times \mathbf{P}^1$ with $g(C) \geq 1$, the full-even-bulk calculation in Lemma 3.3 has two even-rank-two factors, each with centered operator $(2-2g(C))h$. For genus one it is identically zero; for larger genus the modified residue discriminant is zero. All even classes here are Tate, so this is also the fixed-base calculation. The horizontal Novikov parameter remains an independent spectator. For $C = \mathbf{P}^1$ the generic factors are simple. Every ruled surface is birational to such a product; surface factorization uses point centers only, whose contributions have already been shown to vanish. The $\mathbf{P}^2$ calculation and point blowups complete the surface classification argument. No fourfold invariance was used to prove these vanishings.

Projective weak factorization of fourfolds now uses only smooth projective centers of dimension at most two. Every correction is zero, proving the last assertion. □

7.4 Endpoint recovery and the proof of the theorem

We first explain why continuation preserves the information just retained. On a connected fixed-base germ an equivariant projector has constant representation multiplicities. Indeed, write the full fiber as a sum $\bigoplus_i V_i \otimes M_i$ of irreducible $G_{\mathbf{C}}$-representations and multiplicity spaces. The projector acts by the identity on V_i and an idempotent on M_i. Its image has locally constant rank, hence constant rank on the germ. This argument concerns a separated continued projector; it does not continue a decomposition through a collision of eigenvalues.

For $X \in \mathcal{F}$, all even cohomology is Tate and $H^{\mathrm{odd}}(X) = H^3(X)$. The nine endpoint calculations in Proposition 5.2 give one repeated even primary factor: rank two with nonzero discriminant in index-two degrees $1, 2, 3$ and index-one genera $6, 8$, or rank three in index-one genera $2, 3, 4, 5$. All complements have even rank one.

By Lemma 5.1, the rank-one complements have no odd part. Hence the entire $H^3(X)$ lies in the retained factor, and

$$\mathcal{H}_G(X) = [H^3(X, \mathbf{Q})_{\mathbf{C}}].$$

At the small product point, the genus-zero product formula identifies the full connection and its original lattice for $X \times \mathbf{P}^1$ with the tensor product [30, formula (1)]. The independent fiber Novikov parameter separates the two $\mathbf{P}^1$ eigenvalues and every shifted X cluster. The specialized $\mathbf{P}^1$ Lemma 4.2 continues these clusters in all even bulk directions, including mixed ones: cyclicity and even rank persist, and the rank-two discriminant is unchanged. Tensoring with the regularly separated rank-one factor changes the residue only by a scalar. On the fixed base the equivariant-projector argument above preserves the odd multiplicities. The Tate-shifted second copy has the same G-representation. Therefore

$$\mathcal{H}_G(X \times \mathbf{P}^1) = 2[H^3(X, \mathbf{Q})_{\mathbf{C}}].$$

Proof of Theorem 1.3. Choose the common group G for a weak factorization between the two products. Operation and vanishing, followed by endpoint recovery, give

$$2[H^3(X, \mathbf{Q})_{\mathbf{C}}] = 2[H^3(Y, \mathbf{Q})_{\mathbf{C}}] \quad \text{in } R(G_{\mathbf{C}}).$$

Semisimplicity makes this group free on simple classes. Cancel two and obtain an isomorphism of the complexified G-representations.

Put $V = H^3(X, \mathbf{Q})$ and $W = H^3(Y, \mathbf{Q})$. Both have pure weight three. On their Hodge summands the circle records $p - q$, and the common weight records $p + q = 3$; together these determine (p, q). Thus a rational G-equivariant map between V and W is precisely a rational Hodge morphism. Equivalently, no nonzero Tate twist can identify weight-three constituents here.

Finally, equivariant Hom commutes with scalar extension, so

$$\mathrm{Hom}_G(V, W) \otimes_{\mathbf{Q}} \mathbf{C} = \mathrm{Hom}_{G_{\mathbf{C}}}(V_{\mathbf{C}}, W_{\mathbf{C}}).$$

The determinant polynomial on the finite-dimensional rational vector space $\mathrm{Hom}_G(V, W)$ is nonzero because the complexified space contains an isomorphism. Over the infinite field $\mathbf{Q}$, a nonzero polynomial has a rational point where it does not vanish. That point gives the required rational Hodge isomorphism. □

The conclusion retains neither a distinguished integral lattice nor a principal polarization. Those structures are separate inputs in any subsequent Torelli or arithmetic application.

8 A Torelli consequence

Hodge conservation applies to every member of the nine families. Recovering a hypersurface from the resulting rational Hodge structure uses a separate generic Torelli theorem, and its very-general qualification.

Corollary 8.1 (Very-general cancellation)**.** *Let $X \subset \mathbf{P}^4$ be a very general smooth cubic or quartic threefold, and let $Y \subset \mathbf{P}^4$ be any smooth hypersurface of the same degree. Then*

$$X \times \mathbf{P}^1 \textit{ is birational to } Y \times \mathbf{P}^1 \quad \Longleftrightarrow \quad X \cong Y.$$

Proof. Theorem 1.3 identifies the rational third Hodge structures. Here H^3 is the entire primitive middle cohomology. Voisin's rational generic Torelli theorem applies to both degrees [38, Theorem 0.2 and Remarks 0.1, 0.3], and gives $X \cong Y$. The reverse implication is immediate. □

The arithmetic consequence in Appendix E uses Hodge conservation with bounded-degree isogeny and polarization-finiteness theorems. Neither application strengthens the integral or polarized content of Theorem 1.3.

A General bundle formulas and additive invariants

The results in this appendix use the general projective-bundle theorem of Iritani–Koto. For product computations, the cubic proof and the classification and Hodge theorems use only Lemma 4.2 and the ruled-product calculation.

Coefficient fields. The intrinsic ample-adic completion and the Laurent neighborhood of a fibre or exceptional cusp need not map to one another. Iritani's display (1.1) supplies the required common coefficient maps [17, (1.1)]; injectivity for each center summand is proved in Lemma 3.1. For projective bundles the common source is $\mathbf{C}[z, q][[Q, \widehat{\tau}]]$. On the blowup side the ambient term uses the corresponding graded source. A center term begins with its intrinsic reduced coefficient ring from Remark 2.3, maps to the possibly smaller image ring of Remark 5.6, and is then pulled to a separate target ring for that summand in the external direct sum. Embed the generic fields of the ambient common image and of these separate center images in a fixed algebraically closed overfield and form every primary block there. Base change after this splitting extension preserves the rank, nilpotent Jordan type, regular formal gauge, and formal exponent classes of each block; forming primary factors over a nonsplitting field before extension would not.

Restriction to even cohomology. The comparison is restricted to even cohomology, not merely to the even bulk base. The supermodule conventions and the pull-push and Fourier formulas in Iritani's Section 5.8 and Iritani–Koto's construction are parity-preserving. After the odd bulk variables are set to zero, the comparison and its inverse therefore restrict to mutually inverse maps on H^{ev}. This is the QDM on which the additive invariant is defined.

Regularity in z. The comparisons are regular in z. Iritani's Theorem 5.18 and Iritani–Koto's Theorem 5.1 are respectively defined over rings of the form

$$\mathbf{C}[z]((q_{\mathrm{exc}}^{-1/\mathfrak{s}}))[[Q, \widetilde{\tau}]], \qquad \mathbf{C}[z]((q_{\mathrm{fib}}^{-1/r'}))[[Q, \widehat{\tau}]],$$

where Iritani–Koto take $r' = r$ when $r - 1$ is even and $r' = 2r$ when $r - 1$ is odd. Here $\mathfrak{s} \in \{c-1, 2(c-1)\}$ is Iritani's parity-dependent ramification index from [17, (5.11)]. Their inverses use no negative powers of z; see also [18, Remark 5.3]. The maps are homogeneous but need not have degree zero: the projective-bundle map has degree $-(r-1)$, while Iritani's ambient and center components have degrees 0 and $-c$, respectively [18, Theorem 5.1(3)][17, Theorem 5.18(3)]. The cited maps intertwine the actual z-connections. A common grading normalization can translate the residue by a scalar, preserving its discriminant; no eigenline is independently regraded. The graded completed rings embed coefficientwise in $\overline{K}[[z]]$; no identification with an ordinary tensor-product completion is needed.

Proposition A.1 (Intrinsic formulas for the two counts)**.** *For either $I = I_{\exp}$ or $I = I_{\mathrm{lat}}$, a smooth codimension-c center $Z \subset Y$ and a rank-r vector bundle V satisfy*

$$I(\mathrm{Bl}_Z Y) = I(Y) + (c-1)I(Z), \qquad I(\mathbf{P}_Y(V)) = rI(Y). \tag{A.1}$$

The formulas are componentwise for disconnected centers.

Proof. Projective bundles. For a rank-r bundle, Iritani–Koto's Theorem 5.1 gives r copies of the base QDM after Laurent extension [18, Theorem 5.1]. Its invertible joint bulk Jacobian gives independent unit coordinates on the r target factors. In the j-th factor, changing its unit coordinate adds an independent scalar $u_j I$ to Euler multiplication. For two finite spectra, the resultant of their translated characteristic polynomials is a nonzero polynomial in $u_i - u_j$, with leading coefficient one. Hence the spectra of distinct factors are disjoint at the generic point: their primary blocks do not fuse when the direct sum is formed. Regularity, restriction to even cohomology, the fixed splitting field, and invariance under common scalar grading shifts therefore carry the invariant blockwise across the comparison, and additivity gives (A.1). Tensoring the bundle by a line bundle, when needed for the global-generation convention, changes neither the projective bundle nor the intrinsic Novikov embedding [18, Remark 1.2 and Remark 5.2].

Blowups. For a codimension-c blowup, Iritani's Theorem 5.18 gives one ambient QDM and $c-1$ center QDMs [17, Theorem 5.18]. Parts (4), (6), and (7) give the comparison asymptotics, the leading reconstruction parameters, and the invertible combined bulk Jacobian, respectively. The independent unit coordinates supplied by part (7) give the same nonzero resultant argument, so the spectra of the ambient and individual center summands are pairwise disjoint generically. Regularity, restriction to even cohomology, the fixed splitting field, invariance under common scalar grading shifts, and Lemma 3.1 for each center copy then give (A.1). Faithful center base change identifies each of the $c-1$ contributions with $I(Z)$. Each comparison is evaluated over its own common coefficient ring; the proof never composes Laurent comparisons recursively. □

A.1 The additive formulation

For retained block data $\mathcal{T}$, let $\mathcal{L}_{\mathcal{T}}$ be the monoid of finite multisets of regular-isomorphism classes of blocks, with addition by multiset union. Write $\mathcal{B}_{\mathcal{T}}(Y)$ for the block multiset of Y. An *additive invariant* is a homomorphism

$$w : \mathcal{L}_{\mathcal{T}} \longrightarrow A, \qquad I_w(Y) := w(\mathcal{B}_{\mathcal{T}}(Y)),$$

where A is a commutative monoid, possibly without subtraction. For a blowup $\beta : \mathrm{Bl}_Z Y \to Y$ of codimension $c \geq 2$, retain the data of each center occurrence separately:

$$\omega_j = (\beta, Z, j, \varsigma_j, \chi_j), \qquad 0 \leq j \leq c-2.$$

Here ς_j is the reconstruction parameter and χ_j its numerical Novikov specialization. Set $c_w(\omega_j) := w(\mathcal{B}_{\mathcal{T}}(Z;\omega_j))$; equality of these values need not identify the block multisets.

Theorem A.2 (A birational invariance criterion)**.** *Let $w : \mathcal{L}_{\mathcal{T}} \to A$ be an additive invariant unchanged by the allowed regular scalar extensions and formal coordinate changes. Suppose*

$$I_w(\mathrm{Bl}_Z Y) = I_w(Y) + \sum_{j=0}^{c-2} c_w(\omega_j), \qquad \mathrm{codim}_Y Z = c \geq 2. \tag{A.2}$$

If every summand associated with a smooth center of dimension at most $d-2$ has zero correction, then I_w is a birational invariant of smooth projective d-folds.

Proof. Projective weak factorization writes a birational map of smooth projective d-folds as a zigzag of blowups and blowdowns along smooth centers of codimension at least two [4, Theorem 0.3.1]. By (A.2) and the vanishing of every center correction, the invariant is unchanged at each step; the equalities therefore telescope. Disconnected centers are treated componentwise, while a divisor blowup is an isomorphism. □

For the quantum comparisons one also has the bundle formula

$$I_w(\mathbf{P}_Y(V)) = rI_w(Y), \qquad \mathrm{rk}\, V = r. \tag{A.3}$$

It is not a hypothesis of the birational invariance criterion.

Proposition A.3 (Marker formulas for the QDM comparisons)**.** *Let an additive invariant be preserved by the regular scalar extensions and formal coordinate changes described above, and by every recorded grading suspension if its block type retains grading. The projective-bundle and blowup QDM comparisons on even cohomology then supply* (A.3) *and* (A.2)*; the contribution of the j-th center summand is evaluated using its separate data ω_j.*

Proof. The comparison proof of Proposition A.1 uses only regular transport, generic separation and additivity. Apply it to w, retaining each center occurrence and any grading suspensions in its block data. □

Corollary A.4 (Threefold criterion)**.** *Let Y be a smooth projective complex threefold. If $I_{\mathrm{lat}}(Y) > 0$, then $Y \times \mathbf{P}^1$ is irrational, and so is Y. In particular, this holds when $I_{\mathrm{exp}}(Y) > 0$.*

Proof of Corollary A.4. The projective-bundle formula gives

$$I_{\mathrm{lat}}(Y \times \mathbf{P}^1) = 2I_{\mathrm{lat}}(Y) > 0.$$

Every center in projective weak factorization of fourfolds has dimension at most two, so Proposition 3.4 makes every correction zero. The intrinsic blowup formula therefore makes I_{lat} birationally invariant in dimension four. The generic even connection of $\mathbf{P}^4$ has only rank-one primary summands, so $I_{\mathrm{lat}}(\mathbf{P}^4) = 0$. Hence $Y \times \mathbf{P}^1$ is irrational. Rationality of Y would imply rationality of this product. Finally, $I_{\mathrm{exp}} \leq I_{\mathrm{lat}}$ gives the last assertion. □

B Additive and spectral extensions

The blowup formulas also define an additive invariant on the Grothendieck group of varieties. We first retain the squared exponent gaps instead of only counting them. The canonical modified lattice is essential here: arbitrary separate integer shifts of the two exponents can change their squared difference.

Lemma B.1 (A fixed target for the residue spectrum)**.** *For every counted generic block, $\delta^\sharp$ belongs to*

$$\mathcal{D} = \mathbf{C} \setminus \{n^2 : n \in \mathbf{Z}\}.$$

Its value is preserved by the regular comparisons used in the blowup and projective-bundle formulas.

Proof. The finite jet defining the residue is over a finite algebraic extension of the reduced generic coefficient field K_Y: spectral projectors and the finitely many Sylvester equations needed for this jet require only algebraic scalar extension. Proposition 2.5 makes its discriminant constant in every even bulk direction. On the reduced source, a divisor direction acts by $X_d \mapsto (D \cdot d)X_d$, while the other bulk directions are ordinary partial derivatives. Thus the discriminant is killed by all logarithmic derivations in the numerical curve and remaining bulk variables.

The common constants of K_Y under these derivations are $\mathbf{C}$. Here is a coefficient argument that also applies to the completed ring. Write a constant fraction as f/g. Simultaneous formal rescaling of the monomials gives $f(e^t x)g(x) = f(x)g(e^t x)$, by uniqueness of formal Taylor expansion. At each bounded ample-plus-bulk degree only finitely many monomials occur. Their distinct exponential characters are linearly independent, by the Vandermonde argument in Lemma 3.1. Comparing the coefficient of each monomial therefore gives $f_a g_b = g_a f_b$ for every pair of indices a, b. Choose b with $g_b \neq 0$; then $f = (f_b/g_b)g$.

The same constant field holds in an algebraic extension: differentiating the monic minimal polynomial of an algebraic constant gives a polynomial of smaller degree vanishing at it, so every coefficient of the minimal polynomial is constant. Since $\mathbf{C}$ is algebraically closed, the algebraic constant lies in $\mathbf{C}$. The exponent criterion excludes exactly the integer squares.

A regular comparison maps the centered leading operator and its image line to their counterparts. It therefore maps $E^\sharp$ isomorphically to the corresponding modified lattice. On reduction modulo z, the residues are conjugate. A common scalar shift of the residue also leaves $(\operatorname{tr} R)^2 - 4\det R$ unchanged. These are the transformations supplied by the comparisons; no separate change of the two exponent representatives is needed. The faithful coefficient maps fix $\mathbf{C}$, so they preserve the same numerical value in $\mathcal{D}$. □

Put $G_{\mathrm{disc}} = \bigoplus_{\delta\in\mathcal{D}} \mathbf{Z}e_\delta$ and

$$\mathfrak{S}(Y) = \sum_{E \text{ counted}} e_{\delta^\sharp(E)} \in G_{\mathrm{disc}}.$$

Sum over connected components and set $\mathfrak{S}(\varnothing) = 0$. The augmentation $\epsilon(e_\delta) = 1$ recovers I_{exp}. For I_{lat}, replace $\mathcal{D}$ by $\mathcal{D}_{\mathrm{lat}} = \mathbf{C} \setminus \{0\}$ and use the same definitions, written $G_{\mathrm{disc,lat}}$, $\mathfrak{S}_{\mathrm{lat}}$ and ϵ_{lat}. The preceding proof uses exponent separation only to exclude integer squares; with the lattice rule it instead excludes zero. All formulas below therefore apply to both spectra. In particular, for the three families in Theorem 6.1,

$$\mathfrak{S}_{\mathrm{lat}}(V_d \times \mathbf{P}^1) = 2e_{\delta_d}, \qquad (\delta_1, \delta_2, \delta_3) = (16/9, 1, 4/9).$$

These fourfolds are not birational across different degrees. Degrees one and three have the same unordered exponent classes modulo $\mathbf{Z}$, but different canonical discriminants.

Proposition B.2 (Intrinsic operation formulas)**.** *Let Y be smooth projective over $\mathbf{C}$. For a smooth closed center $Z \subset Y$ of codimension $c \geq 1$ and a rank-r vector bundle V on Y,*

$$\mathfrak{S}(\mathrm{Bl}_Z Y) = \mathfrak{S}(Y) + (c-1)\mathfrak{S}(Z), \qquad \mathfrak{S}(\mathbf{P}_Y(V)) = r\mathfrak{S}(Y).$$

The same formulas hold for I_{exp}, and also for $\mathfrak{S}_{\mathrm{lat}}$ and I_{lat}. For centers of differing component dimensions, apply the formula separately to each component.

Proof. Lemma 3.1 identifies every center occurrence with a faithful extension of its intrinsic module in every dimension. The constant-field and regular-lattice argument of Lemma B.1 preserves each discriminant for either exclusion rule. Independent unit coordinates separate the summands generically, as in Proposition A.3. Consequently their spectra add. The projective-bundle comparison gives r copies in the same way. Augmentation proves the scalar formulas. Disjoint centers may be blown up successively, giving the last assertion. □

Theorem B.3 (Additive Grothendieck-group invariants)**.** *There are unique additive homomorphisms*

$$\mathcal{S} : K_0(\mathrm{Var}_{\mathbf{C}})/(\mathbb{L}-1) \longrightarrow G_{\mathrm{disc}}, \qquad \mathcal{I} = \epsilon\mathcal{S} : K_0(\mathrm{Var}_{\mathbf{C}})/(\mathbb{L}-1) \longrightarrow \mathbf{Z}$$

agreeing with $\mathfrak{S}$ and I_{exp} on smooth projective varieties. Here $\mathbb{L} = [\mathbf{A}^1]$, $\mathcal{I}(1) = \mathcal{I}(\mathbb{L}) = 0$, and for a smooth cubic threefold X, $\mathcal{S}([X]) = e_{4/9}$ and $\mathcal{I}([X]) = 1$. The same assertions hold for unique additive extensions $\mathcal{S}_{\mathrm{lat}}$ and $\mathcal{I}_{\mathrm{lat}} = \epsilon_{\mathrm{lat}}\mathcal{S}_{\mathrm{lat}}$ of $\mathfrak{S}_{\mathrm{lat}}$ and I_{lat}, with spectrum target $G_{\mathrm{disc,lat}}$.

Proof. Bittner's presentation of the abelian group uses smooth projective generators, the empty class, and the blowup relation [5, Theorem 3.1]. Since the exceptional divisor is $\mathbf{P}_Z(N_{Z/Y})$, Proposition B.2 gives

$$\mathfrak{S}(\mathrm{Bl}_Z Y) - \mathfrak{S}(\mathbf{P}_Z(N_{Z/Y})) = \mathfrak{S}(Y) - \mathfrak{S}(Z)$$

in G_{disc}. Thus the assignment extends uniquely and additively. Since $[Y \times \mathbf{P}^1] = [Y] + \mathbb{L}[Y]$ and $\mathfrak{S}(Y \times \mathbf{P}^1) = 2\mathfrak{S}(Y)$, we have $\mathcal{S}(\mathbb{L}[Y]) = \mathcal{S}([Y])$ on smooth projective generators. Hence $\mathcal{S}$ factors through $(\mathbb{L} - 1)$, and augmentation gives $\mathcal{I}$. Points have no counted block, while the cubic has the single block of discriminant $4/9$ computed in Section 4.2. The same proof applies to the lattice spectrum. □

These are additive group invariants. In particular, $\mathcal{I}$ is not multiplicative: its value at 1 is zero but its value at a cubic is one. The identity $\mathcal{I}(\mathbb{L}a) = \mathcal{I}(a)$ does not make $\mathcal{I}$ a stable-birational invariant.

For $n \geq 1$, the relation $H^{n+1} = q$ in small quantum cohomology shows that Euler multiplication on $\mathbf{P}^n$ has distinct eigenvalues at $q \neq 0$. It remains semisimple at the generic big point. Thus $\mathfrak{S}(\mathbf{P}^n) = 0$ for every n, including $n = 0$. For example, embed a cubic threefold $X \subset \mathbf{P}^4 \subset \mathbf{P}^5$. Then $\mathcal{I}([\mathrm{Bl}_X \mathbf{P}^5]) = 1$, whereas $\mathcal{I}([\mathbf{P}^5]) = 0$. Thus birational invariance already fails in dimension five.

The dimensional limit is shared by every abelian-group-valued invariant with the global blowup formula $J(\mathrm{Bl}_Z Y) = J(Y) + (c-1)J(Z)$. If such an invariant were birationally invariant in dimension $n+2$, blowing up $X \times \mathbf{P}^2$ along $X \times \{p\}$ would give $J(X \times \mathbf{P}^2) = J(X \times \mathbf{P}^2) + J(X)$, hence $J(X) = 0$ for every smooth projective n-fold. This argument requires no projective-bundle formula. Adding further block types while retaining that architecture cannot give an obstruction after two stabilizations.

Proposition B.4 (Equal Hodge diamonds, different invariant values)**.** *Let X be a smooth cubic threefold and $p \in X$. Let $C \subset \mathbf{P}^1 \times \mathbf{P}^1 \subset \mathbf{P}^3$ be a smooth curve of bidegree $(2, 6)$. Then $\mathrm{Bl}_p X$ and $\mathrm{Bl}_C \mathbf{P}^3$ have the same Hodge diamond, but their invariant values are respectively one and zero. Consequently $\mathcal{I}$ does not factor through the Hodge–Deligne polynomial.*

Proof. The linear system $\mathcal{O}(2, 6)$ contains smooth members, and adjunction gives genus $(2-1)(6-1) = 5$. For the cubic, $c(T_X) = (1+H)^5/(1+3H)$ gives Euler characteristic -6. Weak Lefschetz and $K_X = \mathcal{O}_X(-2)$ then give $h^{1,1} = h^{2,2} = 1$, $h^{2,1} = h^{1,2} = 5$, and $h^{3,0} = h^{0,3} = 0$. The Hodge blowup formula shows that both displayed blowups have, besides $h^{0,0} = h^{3,3} = 1$, precisely

$$h^{1,1} = h^{2,2} = 2, \qquad h^{2,1} = h^{1,2} = 5.$$

Their Hodge–Deligne polynomials therefore agree. The intrinsic blowup formula adds only point or curve contributions, all zero, so their invariant values equal those of X and $\mathbf{P}^3$, namely one and zero. □

Corollary B.5 (Contributions in any rationalizing factorization)**.** *Let X be a smooth cubic threefold. In any projective weak factorization of a rationalization $X \times \mathbf{P}^m \dashrightarrow \mathbf{P}^{m+3}$, orient the arrows from the product to projective space and take connected centers. Then*

$$\sum_{\text{blowdowns}} (c_Z - 1)\mathfrak{S}(Z) - \sum_{\text{blowups}} (c_Z - 1)\mathfrak{S}(Z) = (m+1)e_{4/9}.$$

For $m = 2$, only threefold centers of codimension two can contribute; their net spectrum is $3e_{4/9}$, and their net contribution to I_{exp} is three.

Proof. The initial spectrum is $(m+1)e_{4/9}$; projective space has no counted blocks and hence spectrum zero. Telescope the intrinsic blowup formulas. In dimension five, every nontrivial center has dimension at most three, and centers of dimension at most two contribute zero. The remaining centers have codimension two. Applying ϵ gives the scalar identity. This counts contributions with multiplicity, not distinct centers, and does not identify a center as a cubic threefold. $\square$

Limits of the construction. Smooth complex cubic threefolds form one connected deformation family. Unmarked deformation-invariant Gromov–Witten data are consequently the same throughout that family, whereas it contains both stably rational cubics [26] and very general stably irrational cubics [13]. Increasing the collection of such quantum data cannot by itself completely detect stable rationality within the family.

The rank-two pairing argument also has a specific boundary. In rank three, let E_{ij} denote the matrix units and put

$$N = E_{12} + E_{23}, \qquad P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad A_0 = E_{21} - E_{32}.$$

Then

$$N^T P = PN = E_{23} + E_{32}, \qquad PA_0 = E_{21} - E_{12} = -A_0^T P.$$

Thus the constant pairing is horizontal for $A(z) = N/z + A_0$, but $A_0 e_1 = e_2 \notin \ker N$. Pairing horizontality alone does not preserve the kernel flag. For a cyclic Jordan leading term and $S = \mathrm{diag}(1, z, \ldots, z^{r-1})$, regularity of the modified connection requires $(A_k)_{ij} = 0$ whenever $i - j > k$, for $k \geq 0$. Additional geometric filtration compatibility would therefore be needed for that higher-rank modification.

The rank-three odd-dimension selector in Section 5 uses cyclic persistence and parity, and requires no such logarithmic modification.

C An independent split-basis cubic calculation

The notation U_X, D_X is as in (4.3); put $r = (3q)^{1/2}$. This calculation splits the two nonzero eigenlines as well as the zero-primary block and checks the rational calculation independently.

Proposition C.1 (Small-even cubic block reduction)**.** *The characteristic and minimal polynomials of U_X are both $T^2(T^2 - 108q)$. After adjoining r, its eigenvalues are $6r, -6r, 0, 0$, and the zero generalized eigenspace is one rank-two Jordan block. A normalized gauge regular in z gives*

$$\begin{aligned} z^2 \partial_z \widetilde{y}_+ &= (6r + O(z^2))\widetilde{y}_+, \\ z^2 \partial_z \widetilde{y}_- &= (-6r + O(z^2))\widetilde{y}_-, \\ z^2 \partial_z \begin{pmatrix} \widetilde{y}_3 \\ \widetilde{y}_4 \end{pmatrix} &= [N + zA_0 + z^2 A_1 + O(z^3)] \begin{pmatrix} \widetilde{y}_3 \\ \widetilde{y}_4 \end{pmatrix}, \end{aligned} \tag{C.1}$$

where

$$N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad A_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}. \tag{C.2}$$

Proof. The matrix

$$C = \begin{pmatrix} 6r^3 & -6r^3 & 0 & -7r^2 \\ 7r^2 & 7r^2 & -2r^2 & 0 \\ 3r & -3r & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \qquad \det C = -486r^5,$$

puts U_X into

$$C^{-1}U_XC = \begin{pmatrix} 6r & 0 & 0 & 0 \\ 0 & -6r & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Partition as $1 \mid 1 \mid 2$. At each positive order in z, the off-diagonal gauge coefficient is the unique solution of a Sylvester equation between blocks with disjoint scalar spectra. The gauge and inverse are regular in z; explicitly, it has the normalized form

$$G(z) = I + \sum_{n\geq 1} G_n z^n, \qquad G_n \text{ block-off-diagonal.} \tag{C.3}$$

To extract the coefficient used in the residue, write $y = CG(z)\widetilde{y}$, $J = C^{-1}U_XC$ and $B = C^{-1}D_XC$. Label the blocks by $+, -, 0$. If the transformed coefficient is $J + z\widehat{A}_0 + z^2\widehat{A}_1 + \cdots$, then

$$(J + zB)G - z^2G' = G(J + z\widehat{A}_0 + z^2\widehat{A}_1 + \cdots).$$

Thus $(\widehat{A}_0)_{ii} = B_{ii}$, while the first Sylvester equations are

$$J_i(G_1)_{ij} - (G_1)_{ij}J_j = -B_{ij} \qquad (i \neq j).$$

At the next order,

$$\widehat{A}_1 = [J, G_2] + BG_1 - G_1 - G_1\widehat{A}_0.$$

Since G_1, G_2 are block off-diagonal and $\widehat{A}_0$ is block diagonal, the zero block is simply

$$A_1 = (\widehat{A}_1)_{00} = B_{0+}(G_1)_{+0} + B_{0-}(G_1)_{-0} = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}.$$

This computes the needed coefficient without solving for G_2, and proves (C.2). □

D Finite Fano data and their geometric scope

The numerical classification uses nine small matrices and eight classical rationality assertions. The matrix calculation establishes the spectral conditions; the cited family identifications and smooth deformation bridges establish their every-member scope.

Task	Evidence required
Check a supplied matrix	Exact characteristic polynomial, cyclic frame, regular gauge jet and residue calculation.
Geometric identification	Geometric quantum formulas; period reconstruction with its nonvanishing condition; smooth deformation bridges.

D.1 Finite algebra: matrices and residues

In the classical anticanonical-power basis, up to a scalar Euler shift, write the matrix at Novikov parameter one as

$$U = \begin{pmatrix} a & c & e & f \\ 1 & b & d & e \\ 0 & 1 & b & c \\ 0 & 0 & 1 & a \end{pmatrix}, \qquad D = \tfrac{1}{2}\,\mathrm{diag}(3, 1, -1, -3).$$

The nine inputs are:

Family	a	b	c	d	e	f
g=2	120	744	137520	650016	119681280	21690374400
g=3	24	104	3888	13600	504576	18323712
g=4	12	42	792	2340	43632	793152
g=5	8	24	304	800	9984	121088
g=6	6	16	156	380	3600	33120
g=8	4	9	64	140	924	5936
d=1	0	0	240	1248	0	57600
d=2	0	0	48	160	0	2304
d=3	0	0	24	60	0	576

The columns $(1, U1, U^2 1, U^3 1)$ have determinant one, so the matrix is cyclic. For genera $2, 3, 4, 5$ the characteristic polynomial is $T^3(T - s)$, with $s = 1728, 256, 108, 64$, respectively. For genera $6, 8$ it is $T^2(T^2 - 44T - 16)$ and $T^2(T - 27)(T + 1)$. For degrees $1, 2, 3$ it is $T^2(T^2 - s)$, with $s = 1728, 256, 108$. Thus the displayed repeated factors are whole primary factors. Restoring the anticanonical-degree parameter t gives $tB(t)UB(t)^{-1}$, where $B(t) = \mathrm{diag}(1, t^{-1}, t^{-2}, t^{-3})$; in index two the integral Novikov variable is $q = t^2$. This homogeneous substitution, after inverting t, preserves primary ranks and the canonical residue discriminant. A scalar Euler shift changes only the removed exponential.

For a repeated rank-two factor, choose a Jordan chain and a complementary invariant frame. In that frame let J be the leading matrix and B the grading coefficient. Solve the off-diagonal Sylvester equation $[J, X] + B_{\mathrm{off}} = 0$ for the first regular gauge coefficient. The next diagonal coefficient is $(BX)_{ii}$: the derivative term $-X$ is off-diagonal. Applying the elementary modification to this finite jet gives exactly the five discriminants in Proposition 5.2. The independent checker constructs the frame by rational elimination; the other checker computes the projector and reduced inverse directly. Neither requires a higher-rank logarithmic lattice.

D.2 Geometric identification for every smooth member

For index-two degrees $1, 2$, the counting matrices are [22, Theorem 2.6.6]; degree 3 is the cubic calculation from Beauville's theorem. For genera $6, 8$, the matrices and scalar shifts are [35, Theorem 6.1.1].

For genera $2, 3, 4, 5$, the geometric period formulas are those in [33, Sections 8–11]. The sextic in $\mathbf{P}(1, 1, 1, 1, 3)$ satisfies the divisibility hypothesis of Proposition D.9 there. The quartic, the $(2, 3)$ intersection and the $(2, 2, 2)$ intersection satisfy the ampleness hypotheses of Corollary D.5. These results identify the normalized quantum period $G(q) = 1 + \sum_{n \geq 2} d_n q^n$. The printed regularized coefficient of degree n is $n! d_n$.

The reconstruction input is [35, Proposition 6.2.2]. For a smooth Picard-rank-one, index-one Fano threefold, its map from the five counting-matrix coefficients to $d_2, \ldots, d_6$ is invertible when

$$-495d_3d_5 + 261d_2d_3^2 - 312d_4d_2^2 + 432d_4^2 + 56d_2^4 \neq 0.$$

For each of the four families this condition holds. The finite check uses $M = U - aI$, with entries $M_{ij}(q) = M_{ij}(1)q^{j-i+1}$ and unit subdiagonal. It solves $q\partial_q y = M(q)y$, $y(0) = (0,0,0,1)^t$, and compares y_3 with G through degree eight. At degree $n > 0$ the recursion is uniquely solvable because $nI - M(0)$ is invertible. The resulting coefficients agree with the cited periods, and the reconstruction criterion identifies M with the geometric counting matrix. The certificate also checks the source's independent low-order formulas [35, Example 5.4]. Thus the finite comparison is used within a stated uniqueness theorem; it does not replace the geometric period formula.

As a further provenance check, these matrices agree entrywise with the reconstructions accompanying Böhning–Graf von Bothmer–Su'a [34, Sections 9–10 and associated data]. If M_0 is a source matrix at parameter one, then $U = D_0^{-1}M_0^{\mathrm{t}}D_0 + aI$. For index one of degree $k = 2g-2$, $D_0 = \mathrm{diag}(1,1,k,k)$; for index two of degree k, $D_0 = \mathrm{diag}(1,2,4k,8k)$ and $a = 0$.

The smooth classification has seventeen deformation families, indexed by index and degree or genus; we use its statement and Hodge numbers as recalled in [36, Introduction and Table 5], which attributes them to Iskovskikh–Prokhorov. The family presentations in [33, Sections 3–17] include the weighted models. Smooth deformation invariance transports the quantum products with the ample generator marked. It does not transport rationality or identify Hodge structures of different members.

Two presentations require an explicit bridge. The special genus-three model and the quartic are smooth intersections of type $(2,4)$ in $\mathbf{P}(1,1,1,1,1,2)$ [33, Section 9]. Perturbing the degree-two equation to have nonzero coefficient of the weight-two coordinate gives a quartic by elimination. Smoothness is open and the smooth parameter locus is connected. For a special Gushel–Mukai threefold, its linear section of the cone passes through the vertex, while its smooth quadric section avoids the vertex [37, Sections 2–3]. Perturb the linear section to miss the vertex. Properness and openness of smoothness give a smooth local family with ordinary general member; ordinary presentations form an open subset of an irreducible parameter space. Thus both special strata have the same small quantum matrix as their ordinary smooth family.

The remaining eight families are rational for every smooth complex member [23, Section 1.1]. No quantum matrix for those controls is a premise of the classification theorem.

D.3 Reproduction and formal boundary

The bundle `verification/fano-matrices/` contains the exact rational inputs and checks by symbolic projectors, rational elimination, period recursion and source comparison. Its README specifies commands, pinned source hashes, finite domains and trust limits. It accompanies the manuscript in the paper repository. The two implementations agree on all seventeen characteristic polynomials and cyclic frames and all nine rank-two residues, including four zero controls. Only the nine detected inputs above are needed here. The computations do not establish the cited Gromov–Witten formulas, classification, deformation invariance or geometric transport. The accompanying Lean artifact is a partial formalization of algebraic fragments and deductions with explicit geometric inputs. Its claim map records the geometric theorems without formal counterparts. A checked deduction does not verify its supplied geometric premises; those are proved in the manuscript or imported by citation. Thus the mathematical proof, exact computations and partial formalization have distinct roles.

E Cubic partners over extensions of bounded degree

Corollary E.1 (Geometric finiteness)**.** *Fix a finitely generated field $K \subset \mathbf{C}$ of characteristic zero, a smooth cubic threefold X/K and a positive integer D. There are finitely many geometric isomorphism classes of smooth cubic threefolds Y/L, where L/K is finite of degree at most D, for which $X_{\overline{K}} \times \mathbf{P}^1$ and $Y_{\overline{K}} \times \mathbf{P}^1$ are birational.*

Proof. Achter's arithmetic intermediate Jacobian gives $A = J(X)/K$ and $J(Y)/L$ [39, Theorem B]. Theorem 1.3 gives an isogeny of their complex abelian varieties. Homomorphisms of abelian varieties do not change under extension of algebraically closed fields, so the isogeny exists over $\overline{K}$. Orr's bounded-degree theorem, in its corrected version, supplies an isogeny

$$A_{\overline{K}} \longrightarrow J(Y)_{\overline{K}}, \qquad \deg \leq c(A,K)D^{\kappa}$$

[40, Theorem 5.1]. The isogeny itself need not be defined over an extension of degree at most D.

Choose an integer $N \geq c(A,K)D^{\kappa}$ and put $M = \operatorname{lcm}(1,\ldots,N)$. Every possible kernel is a subgroup of the finite group $A[M](\overline{K})$. Hence there are finitely many possible quotients A/H. Each quotient has finitely many principal polarizations modulo automorphisms [41, Theorem 1.1]. Cubic Torelli [11] identifies at most one cubic geometric class for each resulting principally polarized intermediate Jacobian. □

The count is of geometric classes. It gives neither twist finiteness over K nor an effective number of partners.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude in proof exploration, literature research, exact computation, formal verification, and drafting. AI-assisted adversarial review identified, in particular, the need to restrict to even cohomology and to distinguish the exponent-class count from the stronger count requiring canonical-lattice preservation. The author checked the arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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